Suggested teaching order for

IB Mathematics AA SL

Teacher: Mr. Leo

Topics

Topic 4: Statistics and Probability	4.1—4.11
Topic 1: Number and Algebra	1.1—1.7
Topic 2: Functions	2.1—2.10
Topic 3: Geometry and Trigonometry	3.1—3.7
Topic 5: Calculus	5.1—5.11

(paragraphs are according to the syllabus components)

Plan

Year 1	Year 2
70% of the syllabus	30% of the syllabus

YEAR 1		
Topic 4	**Statistics and Probability**	**~27 Teaching hours**
4.1	Basic concepts of Statistics	
4.2	Measures of central tendency – Measures of spread	
4.3	Frequency tables – Grouped data	
4.4	Regression	
4.5	Elementary set theory	
4.6	Probability	
4.7	Conditional probability – independent events	
4.8	Tree diagrams	
4.9	Distributions – Discrete random variables	
4.10	Binomial distribution – $B(n,p)$	
4.11	Normal distribution – $N(\mu,\ \sigma)$	
	Midterm (~October/November)	

Topic 1	**Number and Algebra**	**~19 Teaching hours**
1.1	Numbers – rounding – scientific form	

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1.2	Sequences in general – Series	
1.3	Arithmetic sequences	
1.4	Geometric sequences	
1.5	Applications of G.S. – Percentage growth	
1.6	The binomial theorem	
1.7	Deductive proof	

Topic 2	**Functions**	**~ 21 Teaching hours**
2.1	Lines (or linear functions)	
2.2	Quadratics (or quadratic functions)	
2.3	Functions: domain, range and graph	
2.4	Composite functions: $f \circ g$	
2.5	Inverse functions: f^{-1}	
	Revision 1	
	End of term (~January)	
2.6	Transformations of functions	
2.7	Asymptotes	
2.8	The exponential function	
2.9	The logarithmic function	
2.10	Exponential equations	

Topic 3	**Geometry and Trigonometry**	**~ 25 Teaching hours**
3.1	3D geometry	
3.2	Triangles – sine and cosine rules	
3.3	Applications in 3d geometry – navigation	
3.4	The trigonometric circle – arcs and sectors	
3.5	$\sin\theta,\ \cos\theta,\ \tan\theta$ on the unit circle	
	Midterm (~April)	
3.6	Trigonometric identities and equations	
3.7	Trigonometric functions	

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YEAR 2		
Topic 5	**Calculus**	**~ 28 Teaching hours**
5.1	The limit $\lim f(x)$ – the derivative $f'(x)$: a rough idea	
5.2	Derivatives of known functions – rules	
5.3	Tangent line and normal line at some point x_0	
5.4	The chain rule	
5.5	Monotony – max and min	
	Midterm (~October/November)	
5.6	Concavity – points of inflection	
5.7	Optimisation	
5.8	The indefinite integral	
5.9	Integration by substitution	
5.10	The definite integral – areas between curves	
5.11	Kinematics (displacement, velocity, acceleration)	
	Revision 2	
	Mock (~April)	

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